1. (a) Define ethics and explain its importance in personal, professional, and societal contexts with examples. (04)

Ethics is the branch of philosophy that deals with moral principles and values which guide human behavior and help distinguish between right and wrong.

Importance of ethics:

- **Personal context:**
  Ethics shapes individual character and decision-making. For example, being honest in daily life builds trust and self-respect.
- **Professional context:**
  Ethics ensures responsibility, integrity, and fairness in the workplace. For example, a computer engineer must not misuse user data or write malicious software.
- **Societal context:**
  Ethics maintains social harmony and justice. Respecting laws, human rights, and social norms helps build a safe and cooperative society.

Thus, ethics is essential for building trust, accountability, and long-term sustainability in all aspects of life.

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1. (b) What do you mean by Moral Dilemma? If you are a computer engineer, how do you solve a moral dilemma when you face it? (1+3)

A **moral dilemma** is a situation where a person must choose between two or more conflicting moral principles, and any choice results in some ethical compromise.

As a **computer engineer**, moral dilemmas can be solved by:

- Following professional codes of ethics (such as honesty, privacy, and responsibility)
- Analyzing the consequences of each possible action
- Prioritizing public safety, data privacy, and legal compliance
- Consulting senior professionals or ethical guidelines when necessary

For example, if asked to create software that invades user privacy, an ethical engineer should refuse and report the issue.

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1. (c) Briefly explain “digital signature” and “electronic record”. (2+2)

Digital Signature:
A digital signature is a cryptographic technique used to verify the authenticity and integrity of a digital document. It ensures that the document is signed by a valid person and has not been altered.

Electronic Record:
An electronic record is any information generated, sent, received, or stored in digital form, such as emails, online forms, or database entries.

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2. (a) Describe the steps involved in the patenting process. Why is each step critical for securing a patent? (04)

The main steps of the patenting process are:

1. **Idea disclosure and patent search:**
   Ensures the invention is novel and not already patented.
2. **Drafting the patent application:**
   Clearly describes the invention and defines legal protection.
3. **Filing the application:**
   Establishes the priority date for the invention.
4. **Examination and grant:**
   Patent office reviews novelty and usefulness before approval.

Each step is critical because skipping or poorly completing any step can lead to rejection or weak legal protection.

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2. (b) What is a licensing agreement? Briefly describe the key elements and types of licensing agreements. (1+3)

A **licensing agreement** is a legal contract where the owner of intellectual property allows another party to use it under specific conditions.

Key elements:

- Parties involved (licensor and licensee)
- Scope of rights granted
- Duration and royalty/payment terms

Types of licensing agreements:

- Exclusive license
- Non-exclusive license
- Cross-licensing

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2. (c) What is University-Industry Collaboration (UIC)? Discuss its objectives and benefits for both universities and industries. (04)

University-Industry Collaboration (UIC) is a partnership between academic institutions and industries to promote research, innovation, and skill development.

Objectives and benefits:

- Universities gain funding, practical exposure, and research relevance.
- Industries gain access to skilled graduates, research expertise, and innovation.
- Enhances technology transfer and economic growth.

UIC bridges the gap between theory and real-world application.

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3. (a) What is cyber law? Mention the cyber laws in Bangladesh. (02)

Cyber law refers to laws that govern activities related to computers, networks, and the internet.

Cyber laws in Bangladesh include:

- **Digital Security Act**
- Information and Communication Technology (ICT) Act, 2006

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3. (b) What is cybercrime? List five cybercrime examples that commonly occur in Bangladesh using social networks. (03)

Cybercrime is any illegal activity conducted using computers or the internet.

Common cybercrimes in Bangladesh:

1. Facebook account hacking
2. Online fraud and fake pages
3. Cyber bullying
4. Identity theft
5. Spreading false or defamatory content

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3. (c) How can we prevent cybercrimes in Bangladesh? Give your opinions. (04)

Cybercrimes can be prevented by:

- Raising public awareness about online safety
- Using strong passwords and two-factor authentication

- Updating cyber laws and ensuring strict enforcement
- Improving digital literacy and cybersecurity training
- Reporting cybercrimes to authorities promptly

Prevention requires both legal action and responsible user behavior.

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3. (d) Discuss the punishment for illegal access to any critical information infrastructure according to section 17 of Bangladesh Digital Security Act 2018. (03)

According to **Section 17** of the Bangladesh Digital Security Act 2018, illegal access to critical information infrastructure is a serious offense.

Punishment includes:

- Imprisonment up to **5 years** or fine, or both for first offense
- Imprisonment up to **10 years** or higher fine, or both for repeat offense

This strict punishment aims to protect national security and critical digital systems.

4. (a) Differentiate between a Startup, a Spin-off, and an Incubator in the context of technology transfer. Provide one example for each.

Startup:
A startup is a newly created company that aims to develop and commercialize an innovative product or technology. It usually starts independently and focuses on rapid growth.
Example: A software startup developing an AI-based healthcare app.

Spin-off:
A spin-off is a company formed by separating a technology or research outcome from a parent organization such as a university or research lab.
Example: A company created from university research to commercialize a newly invented medical device.

Incubator:
An incubator is an organization that supports startups and spin-offs by providing infrastructure, mentoring, funding guidance, and technical support.
Example: A university technology incubator supporting student-led startups.

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4. (b) Discuss major challenges in technology transfer policy with examples.

Major challenges in technology transfer policy include:

- **Lack of clear intellectual property (IP) ownership:** Disputes may arise between inventors and institutions.
- **Insufficient funding:** Many innovations fail due to lack of commercialization support.

- **Weak industry–academia linkage:** Research may not match industry needs.
- **Regulatory barriers:** Complex approval processes delay technology adoption.

Example: A university-developed technology fails to reach market due to unclear patent ownership and lack of industry collaboration.

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4. (c) Define commercial law. Briefly explain the essential elements of a contract.

Commercial law is the branch of law that governs business activities, trade transactions, and commercial relationships.

Essential elements of a contract:

- **Offer and acceptance:** One party makes an offer and the other accepts it.
- **Lawful consideration:** Something of value must be exchanged.
- **Legal capacity and lawful object:** Parties must be legally competent, and the contract purpose must be legal.

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5. (a) Distinguish between bribe and gift in a professional setting. Discuss their ethical implications through a comparison.

A **bribe** is something offered illegally to influence a decision or gain unfair advantage.
A **gift** is a voluntary item given without expecting improper benefit.

Ethical implications:

- Bribery violates honesty, fairness, and professional integrity.
- Gifts may be ethical if they are transparent, modest, and not linked to decision-making.

Example: Offering money to secure a contract is bribery, while giving a small souvenir during a festival is generally acceptable.

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5. (b) Explain the responsibilities of professionals towards clients, employers, and society. Provide one example for each.

- **Towards clients:** Maintain confidentiality and provide honest service.
  Example: A software engineer protecting client data.
- **Towards employers:** Act loyally and avoid conflicts of interest.
  Example: Not leaking company trade secrets.
- **Towards society:** Ensure work does not harm public welfare.
  Example: Refusing to develop harmful or illegal technology.

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5. (c) Define partnership. Explain the three essential elements of partnership.

A **partnership** is a business arrangement where two or more persons agree to share profits from a lawful business.

Essential elements:

1. **Agreement:** Partnership must arise from an agreement.
2. **Profit sharing:** Profits (and losses) are shared among partners.
3. **Mutual agency:** Each partner can act on behalf of others.

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6. (a) What is Environmental Ethics? Describe the major Environmental Ethical Theories.

Environmental Ethics studies moral relationships between humans and the natural environment.

Major theories:

- **Anthropocentrism:** Nature is valued for human benefit.
- **Biocentrism:** All living beings have intrinsic value.
- **Ecocentrism:** Entire ecosystems, including non-living elements, deserve moral consideration.

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6. (b) Discuss the role of universities, research institutions, and industries in technology transfer with suitable examples.

- **Universities:** Generate knowledge and innovations through research.
  Example: Developing new materials in labs.
- **Research institutions:** Refine and test technologies.
  Example: Government research labs improving agricultural technology.
- **Industries:** Commercialize technologies and bring them to market.
  Example: A company manufacturing products based on university patents.

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6. (c) Describe the roles of Government in Innovation Policy.

The government plays a key role by:

- Funding research and development
- Creating innovation-friendly laws and policies
- Protecting intellectual property rights

- Encouraging collaboration between academia and industry

These actions promote economic growth and technological advancement.

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7. (a) Define workplace harassment and discrimination. Explain two types of each with suitable examples.

Workplace harassment: Unwanted behavior that creates a hostile work environment.
Types:

- **Sexual harassment:** Unwelcome advances or remarks.
- **Verbal harassment:** Insults or threats.

Workplace discrimination: Unfair treatment based on personal characteristics.
Types:

- **Gender discrimination:** Unequal treatment based on gender.
- **Religious discrimination:** Bias due to religious beliefs.

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7. (b) Describe four key responsibilities of employers in ensuring a respectful and safe workplace.

Employers must:

- Provide a harassment-free environment
- Ensure workplace safety and health standards
- Treat employees equally and fairly
- Establish clear complaint and grievance mechanisms

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7. (c) What are the three pillars of sustainability? Explain how a business can implement strategies that support all three simultaneously.

The **three pillars of sustainability** are:

- **Economic:** Profitability and long-term growth
- **Environmental:** Protection of natural resources
- **Social:** Employee welfare and community well-being

A business can support all three by using eco-friendly production methods, paying fair wages, and maintaining profitable operations.

8. (a) What do you know about IT professionals' Ethics and IEB?

IT professionals' ethics refers to the moral principles and standards that guide the behavior of people working in information technology. These ethics ensure responsible use of technology and protect users, organizations, and society.

Key aspects include:

- **Honesty and integrity:** Not creating or using software for illegal purposes.
- **Privacy and confidentiality:** Protecting user data and personal information.
- **Professional competence:** Performing duties responsibly and avoiding negligence.
- **Social responsibility:** Ensuring technology does not harm society.

IEB (Institution of Engineers, Bangladesh) is a professional body that sets ethical standards for engineers. IEB provides a **code of conduct** that engineers must follow, such as maintaining professional integrity, avoiding corruption, respecting laws, and serving public interest. Following IEB ethics helps ensure trust, professionalism, and accountability in engineering practice.

8.(b) Discuss the importance of industrial legislation.

Industrial legislation refers to laws enacted by the government to regulate industries and protect the rights of workers and employers.

Importance of industrial legislation:

- Protects workers' rights related to wages, working hours, safety, and welfare
- Ensures safe and healthy working conditions in factories
- Prevents exploitation of workers, including women and children
- Maintains industrial peace by resolving labor disputes
- Promotes fair employer–employee relationships

Overall, industrial legislation helps create a balanced, safe, and productive industrial environment.

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8. (c) What are the restrictions imposed on the employment of women and young persons in a factory?

Factories impose special restrictions to protect women and young persons from health and safety risks.

Restrictions on employment of women:

- Prohibited from working during night shifts (except where legally permitted)
- Not allowed in hazardous or dangerous operations
- Maternity benefits and health protections must be ensured

Restrictions on employment of young persons (children and adolescents):

- Children below a certain age are not allowed to work in factories
- Adolescents must have a fitness certificate from a medical authority
- Restricted working hours and no night work
- Prohibited from working in dangerous machinery or processes

These restrictions aim to ensure safety, health, and proper development of women and young workers.